

Yu-Chuan Su

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RESEARCH INTERESTS

- Computer Vision — mobile vision, 360°/omni-directional/VR vision, egocentric vision, embodied vision, object detection, activity recognition, video summarization, generative model, image restoration, image editing, personalized image generation, video generation
- Machine Learning — deep learning, multi-modal learning, graph convolutional neural network, spectral domain neural network, diffusion model

EDUCATION

The University of Texas at Austin
Ph.D. in Computer Science

August 2014 - August 2019

- Advisor: Prof. Kristen Grauman
- Thesis: Learning for 360° Video Compression, Recognition, and Display
- GPA: 3.92 (0~4 scale)

National Taiwan University
M.S. in Computer Science

September 2012 - June 2014

- Advisor: Prof. Winston H. Hsu
- Thesis: Large Scale Mobile Visual Recognition
- GPA: 4.27 (0~4.3 scale)

National Taiwan University
B.S. in Computer Science and Physics

September 2006 - June 2011

- GPA: 91.11 (0~100 scale)

AWARDS AND FELLOWSHIPS

- **Google PhD Fellowship 2017**
- **Best Application Paper Award, ACCV 2016**
- **Best Thesis Award, Chinese Image Processing and Pattern Recognition Society 2015**
- **Calhoun Graduate Excellence Fellowship 2014**
- **KDD Cup 2013**
 - Author-Paper Identification Challenge (Track 1) – **1st place** (in 561 teams)
 - Author Disambiguation Challenge (Track 2) – **1st place** (in 241 teams)
- **College Student Research Training Fellowship**
 - Fellowship for B.S. research from National Science Council, Taiwan
- **Academic Achievement Award, National Taiwan University**
 - President's Award 4 times (top 5% academic performance in semester)
 - Dean's Award (top 10% academic performance at graduation)

SELECTED PUBLICATIONS

- Hsin-Ping Huang, Xinyi Wang, Yonatan Bitton, Hagai Taitelbaum, Gaurav Singh Tomar, Ming-Wei Chang, Xuhui Jia, Kelvin CK Chan, Hexiang Hu, Yu-Chuan Su, Ming-Hsuan Yang
Kitten: A knowledge-intensive evaluation of image generation on visual entities
arXiv 2025
- Nanye Ma, Shangyuan Tong, Haolin Jia, Hexiang Hu, Yu-Chuan Su, Mingda Zhang, Xuan Yang, Yandong Li, Tommi Jaakkola, Xuhui Jia, Saining Xie
Scaling Inference Time Compute for Diffusion Models
Conference on Computer Vision and Pattern Recognition (CVPR) 2025
- Hsin-Ping Huang, Yu-Chuan Su, Ming-Hsuan Yang
Generating Long-Take Videos via Effective Keyframes and Guidance
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025
- Hsin-Ping Huang, Yu-Chuan Su, Deqing Sun, Lu Jiang, Xuhui Jia, Yukun Zhu, Ming-Hsuan Yang
Fine-grained Controllable Video Generation via Object Appearance and Context
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025
- Hexiang Hu*, Kelvin C.K. Chan*, Yu-Chuan Su*, Wenhui Chen*, Yandong Li, Kihyuk Sohn, Yang Zhao, Xue Ben, Boqing Gong, William Cohen, Ming-Wei Chang, Xuhui Jia
Instruct-Imagen: Image Generation with Multi-modal Instruction
Conference on Computer Vision and Pattern Recognition (CVPR) 2024 (Oral)
- Yu-Chuan Su, Kelvin CK Chan, Yandong Li, Yang Zhao, Han Zhang, Boqing Gong, Huisheng Wang, Xuhui Jia
Identity Encoder for Personalized Diffusion
arXiv 2023
- Xuhui Jia, Yang Zhao, Kelvin CK Chan, Yandong Li, Han Zhang, Boqing Gong, Tingbo Hou, Huisheng Wang, Yu-Chuan Su
Taming Encoder for Zero Fine-tuning Image Customization with Text-to-Image Diffusion Models
arXiv 2023
- Yang Zhao, Tingbo Hou, Yu-Chuan Su, Xuhui Jia, Matthias Grundmann **Towards Authentic Face Restoration with Iterative Diffusion Models and Beyond**
International Conference on Computer Vision (ICCV) 2023
- Yang Zhao, Yu-Chuan Su, Chun-Te Chu, Yandong Li, Marius Renn, Yukun Zhu, Changyou Chen, Xuhui Jia
Rethinking Deep Face Restoration
Conference on Computer Vision and Pattern Recognition (CVPR) 2022
- Yu-Chuan Su, Soravit Changpinyo, Xiangning Chen, Sathish Thoppay, Cho-Jui Hsieh, Lior Shapira, Radu Soricut, Hartwig Adam, Matthew Brown, Ming-Hsuan Yang, Boqing Gong
2.5D Visual Relationship Detection
Computer Vision and Image Understanding, 2022
- Yu-Chuan Su, Raviteja Vemulapalli, Ben Weiss, Chun-Te Chu, Philip Andrew Mansfield, Lior Shapira, Colvin Pitts
Camera View Adjustment Prediction for Improving Image Composition
arXiv, 2021
- Yu-Chuan Su, Kristen Grauman
Kernel Transformer Networks for Compact Spherical Convolution
Conference on Computer Vision and Pattern Recognition (CVPR) 2019

- Yu-Chuan Su, Kristen Grauman
Learning Compressible 360° Video Isomers
Conference on Computer Vision and Pattern Recognition (CVPR) 2018
- Yu-Chuan Su, Kristen Grauman
Learning Spherical Convolution for Fast Features from 360° Imagery
Advances in Neural Information Processing Systems (NeurIPS) 2017
- Yu-Chuan Su, Kristen Grauman
Making 360° Video Watchable in 2D: Learning Videography for Click Free Viewing
Conference on Computer Vision and Pattern Recognition (CVPR) 2017 (Spotlight)
- Yu-Chuan Su, Dinesh Jayaraman, Kristen Grauman
Pano2Vid: Automatic Cinematography for Watching 360° Videos
Asian Conference on Computer Vision (ACCV) 2016 (Oral, Best Application Award)
- Yu-Chuan Su, Kristen Grauman
Detecting Engagement in Egocentric Video
European Conference on Computer Vision (ECCV) 2016 (Oral)
- Yu-Chuan Su, Kristen Grauman
Leaving Some Stones Unturned: Dynamic Feature Prioritization for Activity Detection in Streaming Video
European Conference on Computer Vision (ECCV) 2016
- Yu-Chuan Su, Tzu-Hsuan Chiu, Yan-Ying Chen, Chun-Yen Yeh, Winston H. Hsu
Enabling Low Bitrate Mobile Visual Recognition – A Performance versus Bandwidth Evaluation
ACM Multimedia 2013 (Oral)
- Chen-Wei Tsai, Yu-Chuan Su, Guan-De Li, Jeng-Da Chai
Assessment of Density Functionals with Correct Asymptotic Behavior
Physical Chemistry Chemical Physics 2013

RESEARCH EXPERIENCE

With Prof. Kristen Grauman

Graduate Research Assistant

August 2014 - August 2019

Computer Science Department, UT Austin

- Research in computer vision and machine learning
- Attention analysis in ego-centric video
- Feature triage in streaming activity detection
- Vision in 360° videos, including compression, object detection, and automatic cinematography

With Prof. Winston H. Hsu

Masters Student

July 2012 - June 2014

Computer Science Department, NTU

- Research in multimedia analysis and machine learning
- Mobile-friendly visual recognition
- Deep learning for video event detection using transfer learning

With Prof. Winston H. Hsu

Undergraduate Researcher

July 2010 - August 2011

Computer Science Department, NTU

- Research in multimedia analysis and retrieval
- Video question answering and event detection on mobile devices
- Investigate the properties of gaussian like image representations

With Prof. Jeng-Da Chai

Undergraduate Researcher

August 2009 - August 2011

Physics Department, NTU

- Research in Density Functional Theory and Time Dependent Density Functional Theory
- Develop new long-range corrected functionals using laplacian correction
- Study the properties of different long-range correction schemes
- Implement LB94 model potential on Q-Chem

PROFESSIONAL ACTIVITIES

Invited Talks

- Embedded Vision Summit, Santa Clara, May 2023
- Program Office of AI Research, National Tsing Hua University, January 2018
- Graduate Seminar, National Taiwan University, December 2017
- 6th Workshop on Intelligent Cinematography and Editing, Lyon, April 2017
- Vision and Learning Meet-Up: Recent Advances and Experience Sharing from Overseas Taiwanese Scholars, Academia Sinica, January 2017

Organizing Committee

- ICCV Workshop on 360 Perception and Interaction, 2019
- ECCV Workshop on 360 Perception and Interaction, 2018

Area Chair / Senior Program Chair

- Computer Vision and Pattern Recognition (CVPR) 2023, 2024, 2025
- International Conference on Learning Representations (ICLR) 2024, 2025
- Assoc. for the Advancement of Artificial Intelligence (AAAI) 2023, 2024
- Neural Information Processing Systems (NeurIPS) 2024, 2025
- International Conference on Machine Learning (ICML) 2025

Journal Reviewer

- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)
- IEEE Transactions on Multimedia (TMM)
- IEEE Transactions on Graphics (TOG)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Visualization and Computer Graphics (TVCG)
- IEEE Transactions on Human Machine Systems (THMS)
- Computer Vision and Image Understanding (CVIU)
- Data Mining and Knowledge Discovery (DAMI)
- Transactions on Machine Learning Research (TMLR)

Conference Program Committee

- Computer Vision and Pattern Recognition (CVPR) 2018-2022
- International Conference on Computer Vision (ICCV) 2019, 2021, 2023, 2025
- European Conference on Computer Vision (ECCV) 2020, 2022
- International Conference on Machine Learning (ICML) 2019-2021, 2023, 2024
- Neural Information Processing Systems (NeurIPS) 2018, 2019, 2022, 2023
- SIGGRAPH 2021
- SIGGRAPH Asia 2017, 2018, 2021

- International Conference on Learning Representations (ICLR) 2021, 2022
- Assoc. for the Advancement of Artificial Intelligence (AAAI) 2022
- Asian Conference on Computer Vision (ACCV) 2018, 2020
- Winter Conference on Applications of Computer Vision (WACV) 2017, 2018, 2022
- ACM Symposium on User Interface Software and Technology (UIST) 2017, 2021
- International Conference on Robotics and Automation (ICRA) 2020

WORK EXPERIENCE

Moonvalley

April 2025 - Present

Principal Research Scientist

- Led the development of controllability in Marey, including but not limited to all capabilities demonstrated: pose transfer, 3D camera positioning, motion transfer, trajectory control, reference images control, sketch to video.

Moonvalley

October 2024 - April 2025

Member of Technical Staff

- Primary developer of Marey, the first commercially safe video generation model.
- Developed the diffusion model and parallel training for Marey.
- Led the development of the diffusion process for both training and inference.

Google

November 2019 - October 2024

Research Scientist

- Conducted research on intelligent photography and generative models, focusing on innovative algorithmic advancements and practical applications.
- Developed and launched image and video unblurring features across multiple Google products, including Photo Unblur in Google Photos, Portrait Restore in Google Meet, and Face Unblur in Pixel Camera. This key enhancement significantly contributed to the Pixel 7 achieving the highest DXO score in 2022. The unblur features were prominently featured in announcements at Google I/O 2022 and Made by Google 2022 events.
- Developed and launched personalized image generation capabilities on Google Cloud (Generative AI on Vertex AI), incorporating both fine-tuning based (DreamBooth) and fine-tuning free methods (Instruct-Imagen, published in CVPR 2024).
- Led the development of the Generative-Selfie feature, enabling fine-tuning free, customized face and person image generation from a single reference image for both Google products and corporate clients.
- Led the development of Google Research's video generation model. The model is based on Gemini 2 with improved training time scalability.
- Developed text-based image editing capabilities for Google's image generation feature, with product delivery currently in progress.

Google

May 2018 - August 2018

Software Engineer Intern

- Work on unsupervised video understanding and depth estimation

Yahoo!

July 2013 - August 2013

Technical Intern

- Work on knowledge graph analysis with Yahoo! search